Data Visualization Proposal

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Abstract

New York City records hundreds of thousands of complaints every year, but counting offences by borough is only a starting point. The questions that matter to decision-makers are where, when and what kind of offence concentrates, and what could be done about it. This project uses the NYPD Complaint Data (2020-2026) to help the Mayor of New York City and the NYPD reduce incidents and make the city safer. The Mayor and the police chief have little time and are not data scientists, so each figure is meant to show one pattern and the decision it supports.

Goals of each hypothesis. Hypothesis 1 studies how reported crime concentrates by borough, time and severity, to support the deployment of patrols. Hypothesis 2 studies where assaults on police and peace officers concentrate beyond what total complaints would suggest, to support officer safety. Hypothesis 3 aims to identify whether underage victim crime spikes at specific times of day can be associated with residence-school-residence commutes, as well as time home after school. Hypothesis 4 studies whether the rise of complaints in the subway since 2020 comes from serious crime or from fare and transit-rule enforcement, to support where violent crime patrols are placed.

Contents of this report. For each hypothesis, we present the research question, the dataset gaps, the transformations applied and planned, the exploratory `ggplot2` figures that led to it (the Extra information section features multiple line of reasoning aid figures, as well as result strengthening plots), the first results and what the Mayor or the NYPD could do with the answers. Each hypothesis will be tested with visualizations in the next checkpoints.

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1 Hypothesis 1: spatial and temporal tactical concentration

Research question

Where and when do reported crimes concentrate across New York City, and how do their severity and offense types shape operational patterns?

Dataset gaps

The dataset holds 3 386 974 complaints reported from January 2020 to June 2026. 16 630 occurred before 2020 (`old_occurrence`), mixing late reports with impossible years (1015, 1900). Offense descriptions (`ofns_desc`, 68 values) are sometimes truncated or duplicated, and 5 835 rows lack a borough. Unknown times are recorded as round hours: 76 513 complaints at 12:00 and 40 142 at 00:00, against about 800 in neighbouring minutes.

Data transformations

`01_clean.R` converted `"(null)"` and invalid age groups to `NA`. We exclude occurrences before 2020, keeping 3 370 344 complaints, aggregated by borough, offense level, type and hour. Expected steps: merging duplicated offense descriptions, flagging round-hour times and normalising borough counts by population.

Exploratory analysis

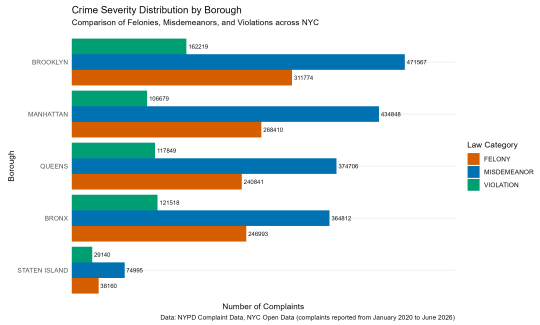


Figure 1: Complaints by offense level and borough. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

Boroughs: Figure 1 shows Brooklyn and Manhattan leading in absolute volume, with misdemeanors dominating everywhere. Raw counts partly reflect population size.

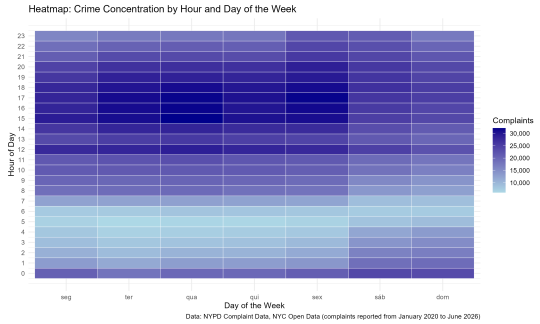


Figure 2: Complaints by hour and weekday (group exploration). Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

Hour and weekday: Figure 2 shows a minimum at 04:00–06:00, peaks on weekday afternoons and high activity on weekend nights, a baseline for patrol scheduling.

Hypothesis

Reported crime in New York City does not distribute randomly; it concentrates in specific boroughs, time windows and contexts. Analysing high-frequency offenses and high-severity incidents, we can provide evidence to optimise NYPD patrol and resource allocation.

First results

Offense types and severity: Figure 3 shows petit larceny (misdemeanor) and harassment (violation) each exceeding 500k complaints, while the most frequent felonies (grand larceny, felony assault, robbery) are far less common.

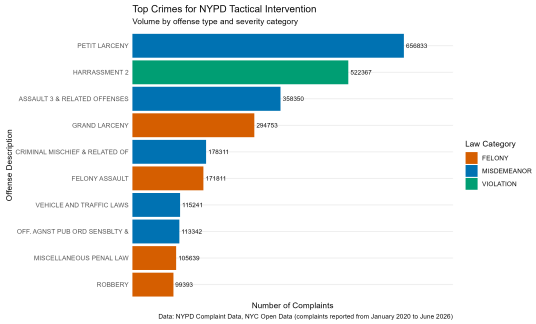


Figure 3: Top offenses coloured by level. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

Time of day and severity: Figure 4 shows the same cycle for all levels: a minimum at 04:00–06:00 and a plateau at 15:00–18:00, felonies peaking at 17:00. Spikes at 00:00 and 12:00 are partly unknown times recorded as round hours.

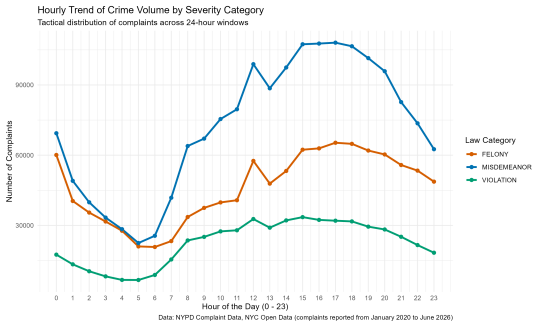


Figure 4: Hourly crime by severity level. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

What the Mayor and the NYPD could do

Reinforce high-visibility patrols in the afternoon and early evening, when every level peaks, and in the highest-volume boroughs once adjusted for population. Prevention campaigns could target petit larceny and harassment, while specialised units focus on felonies such as felony assault and robbery.

2 Hypothesis 2: assaults on police and peace officers

Research question

Where and when are assaults on police and peace officers concentrated?

Line of reasoning

My first idea was to check if adults use children to commit crimes, but the data does not record this. Suspects recorded as under 18 were too few in drug complaints (481 of 73 125 with a known age). With all crimes, 52% had no known suspect age, so I changed direction. In the data quality checks, a table of the most common offence descriptions for each victim code showed that L was mostly linked to assaults on officers. As L only marks these assaults from 2022, I used the offence description instead. The exploratory plots showed that assaults doubled from 2020 to 2025 while all complaints rose by 40% (Figure 12), and that the same precincts led every year (Figure 13). Counts favour busy precincts, so I compared assaults with all complaints (Figure 5). Precinct 41 stood out, but most of its assaults fall under the Department of Correction, so I repeated the test under NYPD jurisdiction only.

Dataset gaps

- **Victim codes:** `vic_sex` also contains D (business or organisation), E (People of the State of New York) and L, which is not in the data dictionary. L marks none of these assaults in 2020 or 2021 and about 80% from 2023.
- **Premises:** 16.9% of these assaults have a missing or "OTHER" premise.
- **Dates:** day 1 of the month has 19% more complaints than the average day of the month, possibly because unknown dates are recorded as the 1st.

Data transformations

Undergone:

- **Assault selection:** complaints described as ASSAULT POLICE/PEACE OFFICER and reported from 2020 to 2025 were selected (11 735 rows), as 2026 ends in June.
- **Premise groups:** street (`STREET`), other or unknown (`OTHER` and missing values), and other known place (all remaining premises).
- **Ratios:** each precinct's share of these assaults divided by its share of all complaints.
- **Hourly comparison:** times at exactly midnight or noon were removed, as they may stand for an unknown time.

Expected:

- Add arrests per precinct from NYPD Arrests Data (NYC Open Data) to check whether high values reflect more police activity.

Hypothesis

Recorded assaults on police and peace officers are concentrated in a small group of precincts, beyond what their total complaints would suggest, and this pattern persists over time.

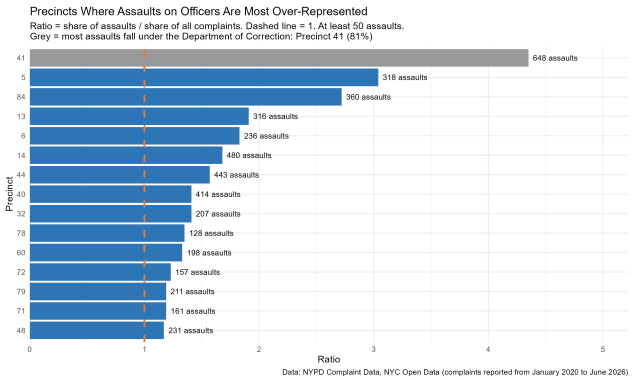


Figure 5: Highest ratios of assaults to all complaints, across all agencies. Grey is Precinct 41, where most assaults fall under the Department of Correction. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

First results

The 10 precincts with the most assaults hold 32.5% of them and 19.3% of all complaints. Precinct 41 has the highest ratio (4.35), but 81% of its assaults (525 of 648) fall under the Department of Correction. Overall, 23.1% of assaults are outside NYPD jurisdiction. Under NYPD jurisdiction only, the top 10 hold 29.3% and 20.1%, and the highest ratios are in Precincts 5, 84, 6, 44 and 32. Precincts 14, 40, 41 and 44 were in the yearly top 10 in all six years. Boroughs differ less, from the Bronx (1.24) to Queens (0.71). Only 36.1% occur on the street, and the main place varies by precinct (Figure 6), the street in 44, transit in 14 and hospitals in 13. Between 7 pm and 5 am there are 46.7% of assaults and 38.4% of all complaints (Figure 14).

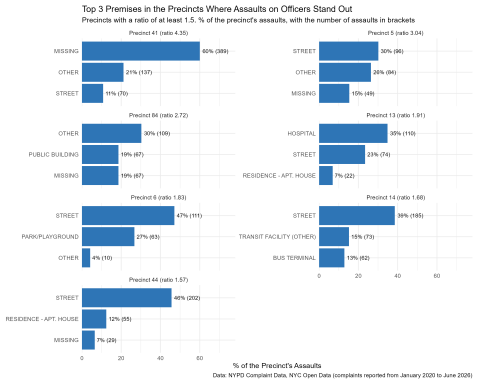


Figure 6: Top 3 premises in precincts with a ratio of at least 1.5, across all agencies. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

What the Mayor and the NYPD could do

The NYPD could review officer safety and support in Precincts 5, 84, 6, 44 and 32, especially at night and around July 4, adapting measures to the main places in each precinct. Precinct 41 should be reviewed with the Department of Correction. These are recorded complaints, so they do not measure the risk per officer, and part of the increase may reflect changes in recording.

3 Hypothesis 3: underage victim spikes at school hours

Research question

Do complaints with victims under 18 (underage) in residences and public transport follow school day commute times and after-school hours?

Line of reasoning

Important exploration (prior ideas, future/expected work) in this week's checkpoint (`02_exploration_rafa`) and Extra information.

When exploring sex vulnerabilities in public housing (3rd dedicated experiment), the only distinct hourly pattern was in underage victims: complaints rose sharply at very specific hours. This encouraged me to explore association with school commute hours.

Dataset gaps

- No population baselines (ratios are shares, not rates)
- Victim race is classified by the officer
- Majority of housing development names are missing
- Missing explicit suspect-victim relationship

Data transformations

Filtering operations (keeping only 2020 to 2025, invalid age groups, non-person sex, missing values, etc.) and derived variables (premises to residence and public transport, flagging underage, weekdays vs weekends, school months vs summer break), with some conversions to percentages within each.

Hypothesis

Complaints with underage victims become sharply more common during specific residence-school-residence commute times (public transport) and after-school time (residences) than at other hours of the day.

First results

In public housing, girls under 18 were the only age group whose complaints dropped during school hours and peaked right after them, on weekdays but not at weekends. To test whether this follows the school day, the analysis was widened to all victims under 18 in residences and public transport. Underage victims are a small share of all victims (7.6% in public transport and 4.6% in residences, against 4.9% across all premises), but on weekdays during school months their share in public transport rises from about 6.0% at most hours to 20.2% at 7h and 25.8% at 15h (Figure 7).

In residences, the weekday share of underage victims spikes at the morning commute (4.9% at 7h, up from 3.3% at 6h), dips during school hours (2.6% at 9h) and more than doubles to 5.7–6.1% between 15h and 20h, after school. Weekends show no dip and similar afternoon levels (Figure 8) 00h and 12h spikes are likely unknown times, and thus not considered.

Overall, commute and after school-hour peaks largely disappear, especially in public transport.

Finally, having established the relationship with school commutes, I explored the specific offenses behind these complaints (Figure 9). Both settings carry

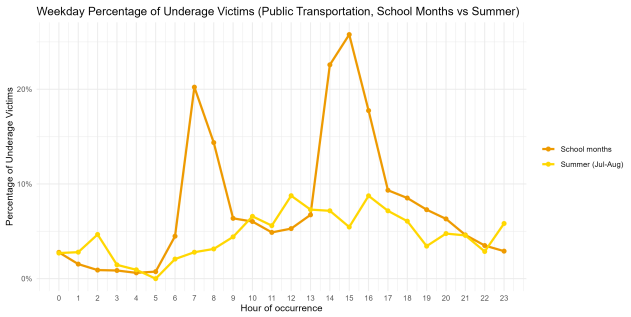


Figure 7: Percentage of victims under 18 in public transport on weekdays, by hour: school months vs summer (July–August), 2020–2025.

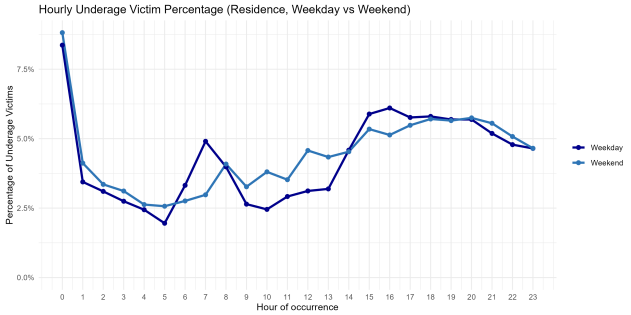


Figure 8: Percentage of victims under 18 in residences, by hour: weekdays vs weekends, 2020–2025.

different risks: In public transport, most common offenses are assault (28%), harassment (21%) and robbery (12%), consistent with risk during the trip itself. In residences, sex crimes are the most common offense (22.2% misdemeanor and 3.3% felony, plus 3.7% rape), followed by harassment (17.7%) and assault (16.4% misdemeanor, 14.4% felony).

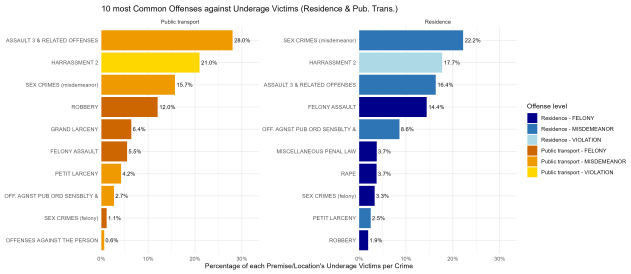


Figure 9: Ten most common offenses against underage victims (2020–2025).

Specifics differ by trip (assault, harassment, robbery), and at home (sex crimes, harassment, assault), leading to the specialized solutions below.

What the Mayor and NYPD could do

- **Schools:** sessions on healthy relationships and sexual education, regular wellbeing checks during mandatory school time, reaching children who may be abused at home (a case the data cannot exclude).
- **Public transport:** uniformed staff at commute hours as a visible deterrent, and undercover staff to protect inside vehicles and upon entry-exiting.
- **After school:** more extracurricular activities, study groups and clubs, with anonymous access to psychologists.

Measuring impact: tracking the new percentage of underage victims at mentioned hours.

4 Hypothesis 4: the rise of complaints in the subway

Research question

Is the rise of complaints in the NYC subway since 2020 a rise in crime, or a rise in police enforcement?

Line of reasoning

Comparing complaints by type of premises, the subway grew ×4.7 from 2020 to 2025 (7 142 to 33 884), more than any other premises (Figure 20). In the subway, the class *other offenses related to theft* went from 16 to 13 455 complaints and explains 50% of the rise (Figure 10). Almost all of it (97%) is *theft of services*, which means using a service without paying, and it is all recorded by the N.Y. Transit Police (Figure 21). This suggests fare evasion. Unlike a robbery or an assault, nobody reports fare evasion to the police, so it is usually recorded only when officers catch someone. Its rise can mean that more people evade the fare or that the police check more often, and the data cannot tell the two apart. In both cases, these complaints are not serious crimes.

Dataset gaps

- **Truncated names:** until 2023 two offence classes were cut at 30 characters (`...RELATED TO THEF`), and written in full from 2024, so counted by name they seem to start in 2024.
- **No ridership:** subway use collapsed in 2020 (COVID) and there are no passenger counts, so part of any rise is a return of passengers.

Data transformations

Undergone:

- **Spelling fix:** truncated `ofns_desc` labels given their full spelling.
- **Subway:** premises `TRANSIT - NYC SUBWAY`, reported 2020–2025 (101 078 rows).
- **Three groups:** *enforcement* (theft of services and `N.Y.C. TRANSIT AUTH. R&R`, the transit rules), *felonies*, and *other misdemeanors and violations*.
- **Stations and hours:** 371 stations ranked by all complaints and by felonies (2024–2025), with exact midnight and noon removed.

Expected:

- Add MTA ridership to compute complaints per passenger.
- Split *other misdemeanors*, as drugs (178 to 2 474), weapons (82 to 1 713) and trespass (22 to 805) may be police-initiated too (Figure 10).

Hypothesis

The rise of complaints in the NYC subway since 2020 comes mostly from fare and transit-rule enforcement, not from serious crime. A ranking of stations by total complaints therefore points to some stations because of enforcement, and hides others where serious crime is high.

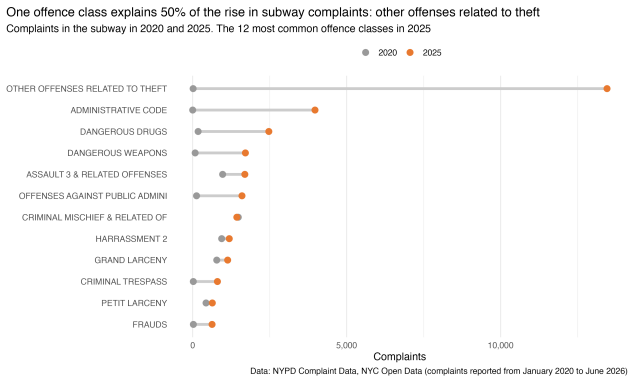


Figure 10: Subway complaints by offence class, 2020 and 2025. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

First results

Enforcement went from 17 complaints in 2020 to 16 934 in 2025, 63% of the rise, while felonies grew ×1.7 (Figure **??**), with felony assault +62%, grand larceny +45% and robbery −31% (Figure 22). In 2024–2025, 5 of the 15 stations with the most felonies are not in the top 15 by all complaints, and 49 Street is 10th by felonies but 74th overall (Figure 11). Enforcement peaks on weekday afternoons, while felonies spread into the night (22% at 00:00–05:59 against 13%) and the weekend (Figure 23). The hypothesis is only partly supported, as other misdemeanors also almost tripled.

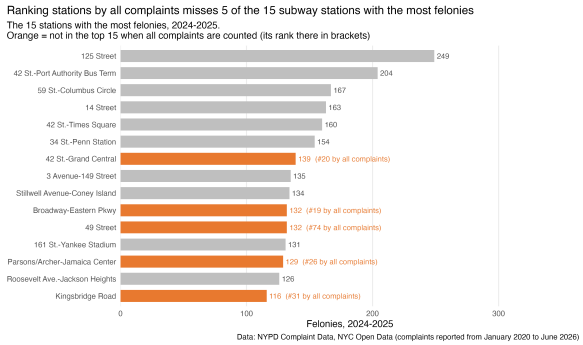


Figure 11: The 15 subway stations with the most felonies, 2024–2025. Orange bars are not in the top 15 by all complaints. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

What the Mayor and the NYPD could do

Read the rise in subway complaints as mostly fare enforcement, not a subway 4.7 times more dangerous. Place violent-crime patrols by felonies, not total complaints (adding 49 Street and Grand Central), at night and at weekends.

A Extra information

A.1 Hypothesis 2

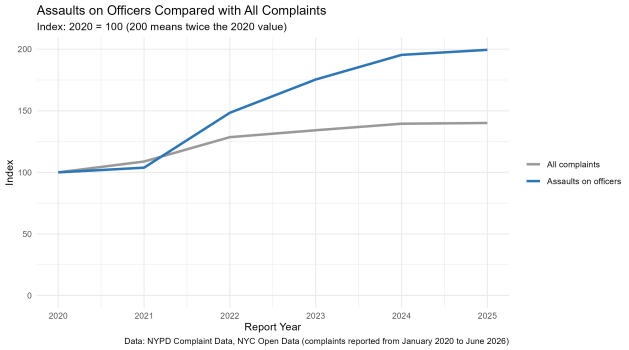


Figure 12: Assaults on officers and all complaints per year, as an index with 2020 = 100. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

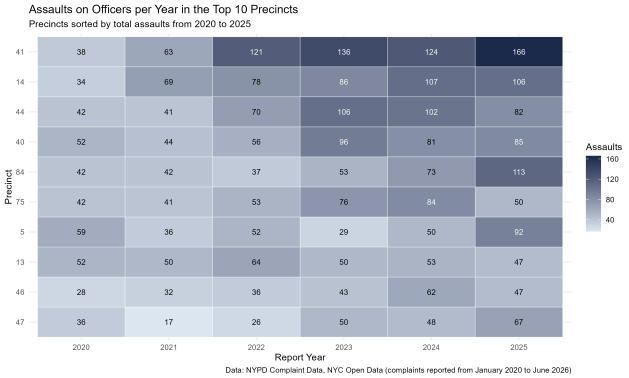


Figure 13: Assaults on officers per year in the 10 precincts with the most assaults. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

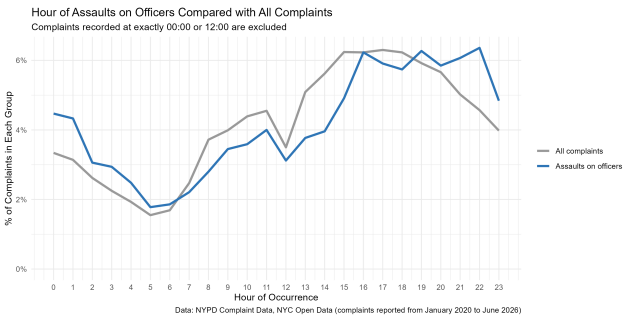


Figure 14: Share of assaults on officers and of all complaints by hour of occurrence. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

A.2 Hypothesis 3

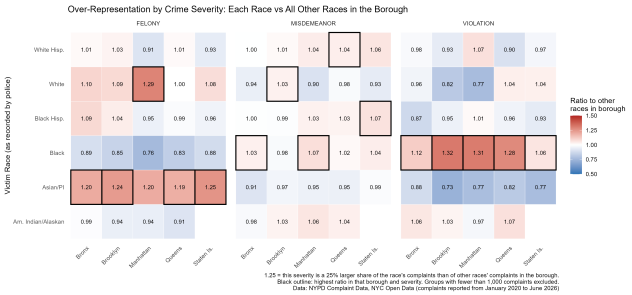


Figure 15: Race: Asian/Pacific Islander victims are over-represented in felonies in every borough (ratios 1.19–1.25), and Black victims in violations (1.06–1.32) while under-represented in felonies (0.76–0.89). Race as recorded by the police; groups under 1,000 complaints excluded.

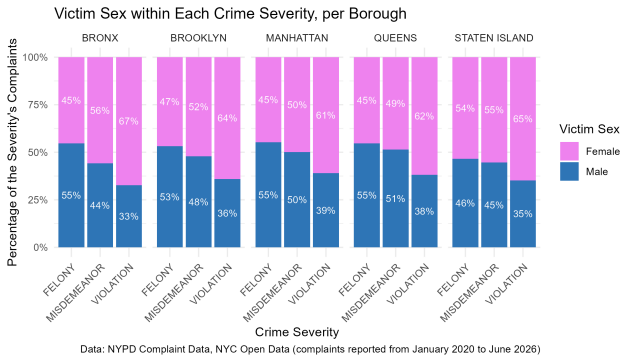


Figure 16: Sex: despite women filing at least as many complaints, the male share of victims rises with severity in every borough (Bronx: 33% of violations, 44% of misdemeanors, 55% of felonies).

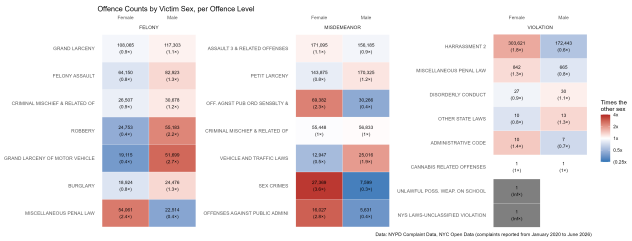


Figure 17: Sex: offense types differ drastically. Sex crimes (3.6×) and harassment (1.8×) mostly affect women; vehicle theft (2.7×) and robbery (2.2×) mostly affect men.

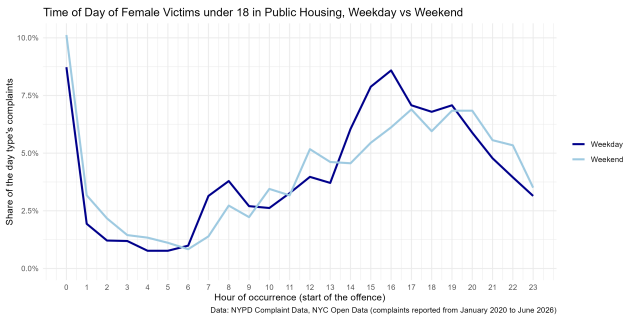


Figure 18: Origin of Hypothesis 3: in public housing, girls under 18 show a sharp afternoon peak (about 8.6% at 16h) on weekdays only; at weekends the profile is flatter and peaks later.

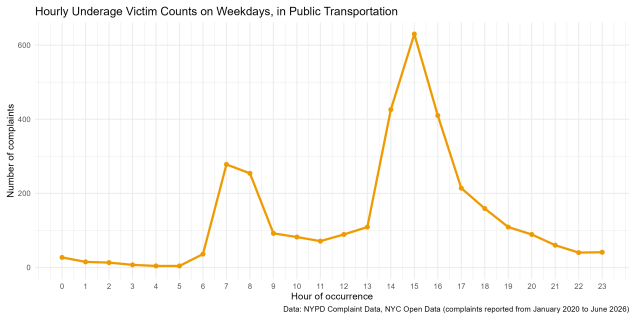


Figure 19: Counts confirm the percentages are not caused by fewer adult victims: complaints with underage victims in public transport peak on weekdays at 7–8h and 15h, with few in between.

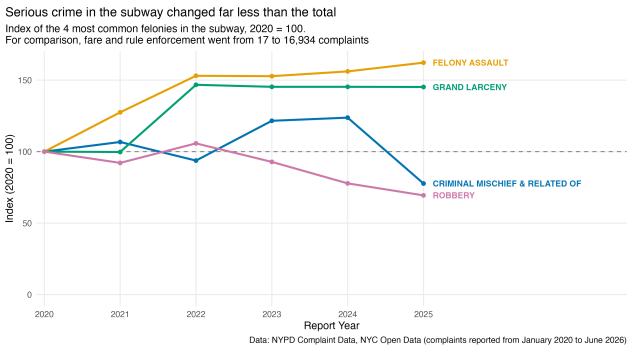


Figure 22: The four most common subway felonies, index with 2020 = 100. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

A.3 Hypothesis 4

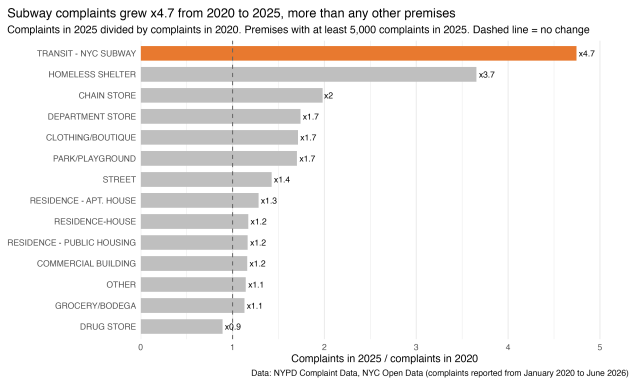


Figure 20: Complaints in 2025 divided by complaints in 2020, by premises with at least 5 000 complaints in 2025. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

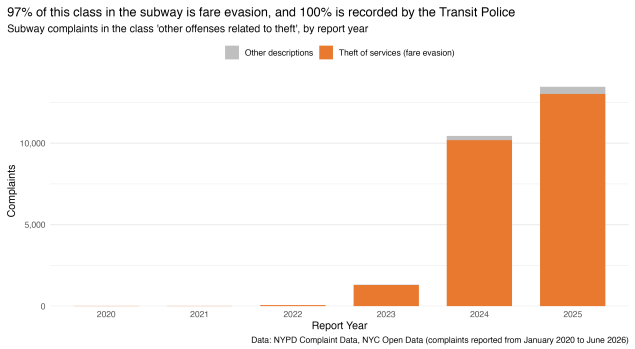


Figure 21: Subway complaints in the class *other offenses related to theft*, by detailed description. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).

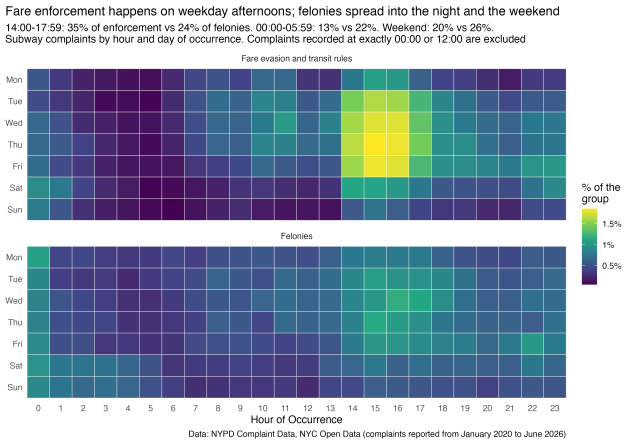


Figure 23: Share of enforcement and of felonies in the subway by hour and weekday. Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026).